

Modernizing Workflows as old as the Railways

Operationalizing Network Automation

AUTOCON S

THE NETWORK AUTOMATION CONFERENCE

Agenda

- Building steam – data
- Speeding up – workflows
- Staying on track – platform
- Refueling – conclusion

Introductions



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Network Automation Architect

DB Systel



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Senior Engineer

Network to Code
on contract with DB Systel

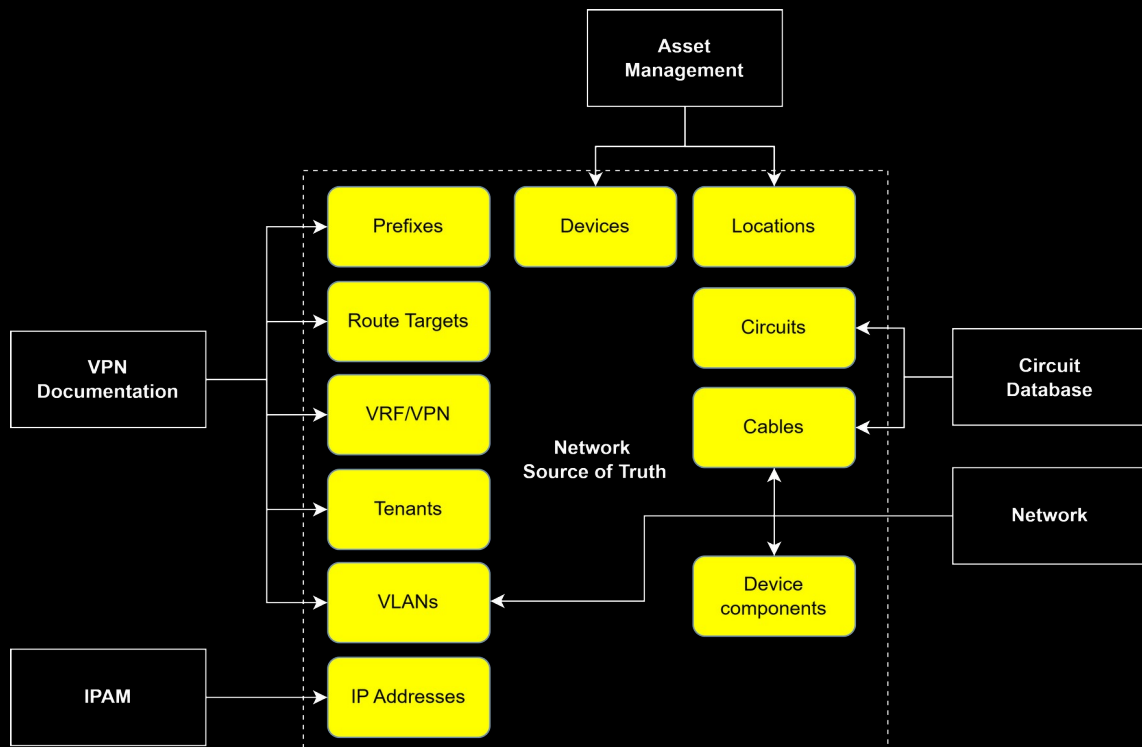
Building steam - data

Where is my Data?

- Siloed legacy systems
- Different Systems of Records
- Multiple responsible teams

Gathering Data

- Single Source of Truth (SSoT) framework
- “We want everything”
- Long path to first automated workflow



The Rise of Data Issues

- Empty locations
- Devices in locations categorized `Not IT infrastructure`
- "Düsseldorf" != "Dusseldorf"
- IPAM inconsistencies in the wild
 - VRF mismatches
 - DNS misconfigurations
 - Prefixes / IP addresses **missing** or with wrong attributes

Acceptance is Difficult

"We are like psychologists right now,
putting all those problems from the past in
people's faces"

- Malte Plescher

First win

- Supportive management
- Uncovered **existing** problems
- "Let's fix it" mentality formed
 - Created daily reports with issues
 - Dedicated clean up team - **superheroes**

+25%

documented IP addresses

Next Steps

- Improved legacy processes
- Engineers started **thinking** "What else can we do?"
- Gave us **momentum**

Speeding up - workflows

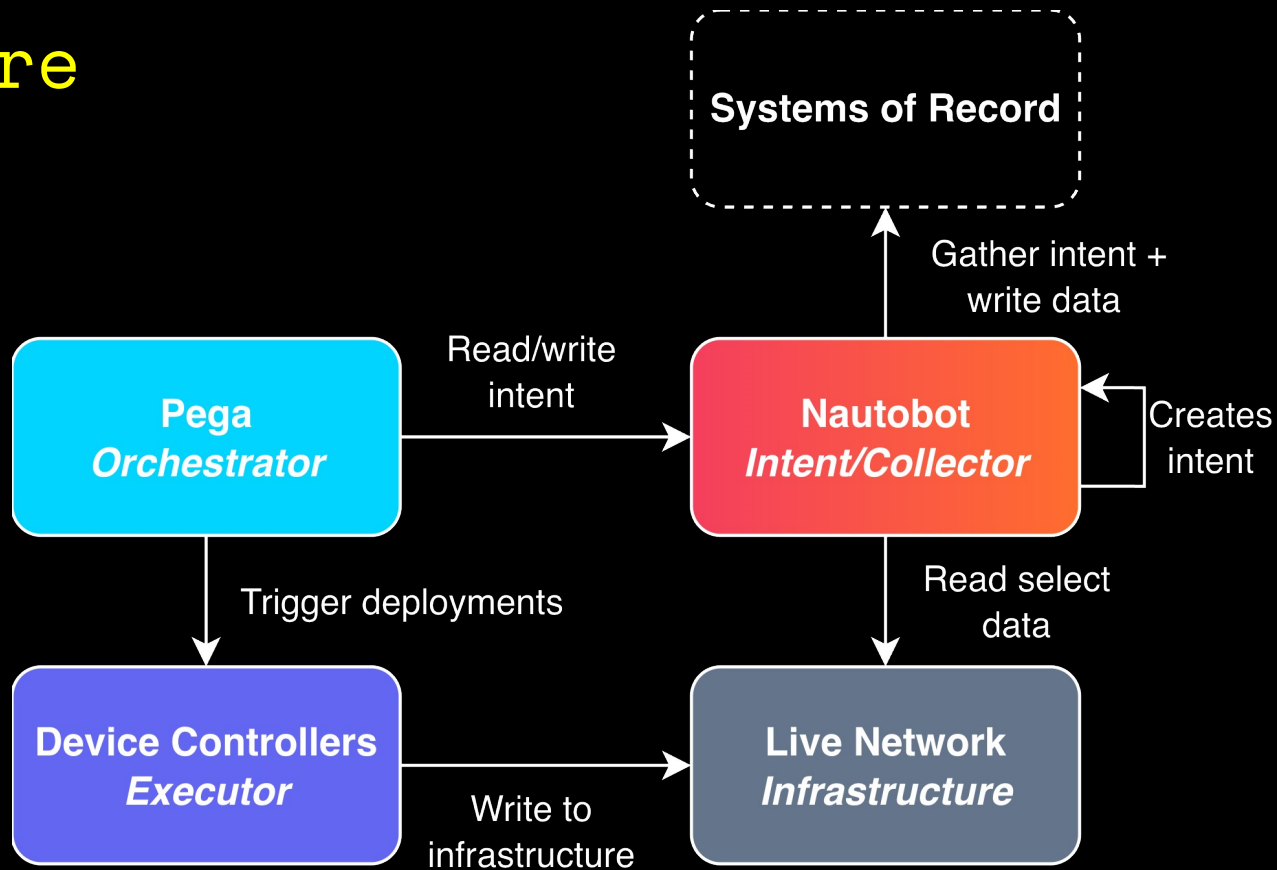


Preparing for workflows

- Heroic efforts are **not** sustainable
- Weak point: data maintenance
- Implemented APIs
 - asset management
 - IPAM

Architecture

*Observability
& presentation
not pictured*



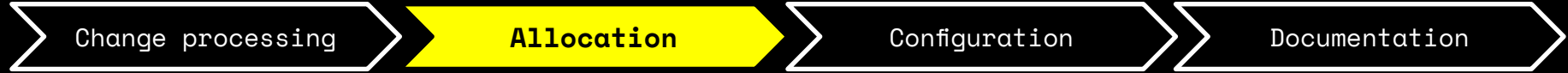
Case Study - VPN Rollout



- Not yet automated
- Manual process in ITSM

10 min == 10 min

Case Study - VPN Rollout



- SoR interaction through small Nautobot jobs
- Decreased effort
- Increased data quality

25 min → 5 min

Case Study - VPN Rollout



- Configuration through NSO service fed by Nautobot job
- Roughly 50 config lines
- Cleaned up brownfield

30 min → 1 min

Case Study - VPN Rollout



- Not yet automated
- Manual process in Visio
- Automated PoC by separate team

5 min == 5 min

70 min → 25 min

Workflow Outcomes

- New skills for engineers
- Production usage of workflows
- Removal of manual steps
- ~50 daily jobs runs in Nautobot

How it was Built

- Initial setup by architects
- Scaling into teams of device owners

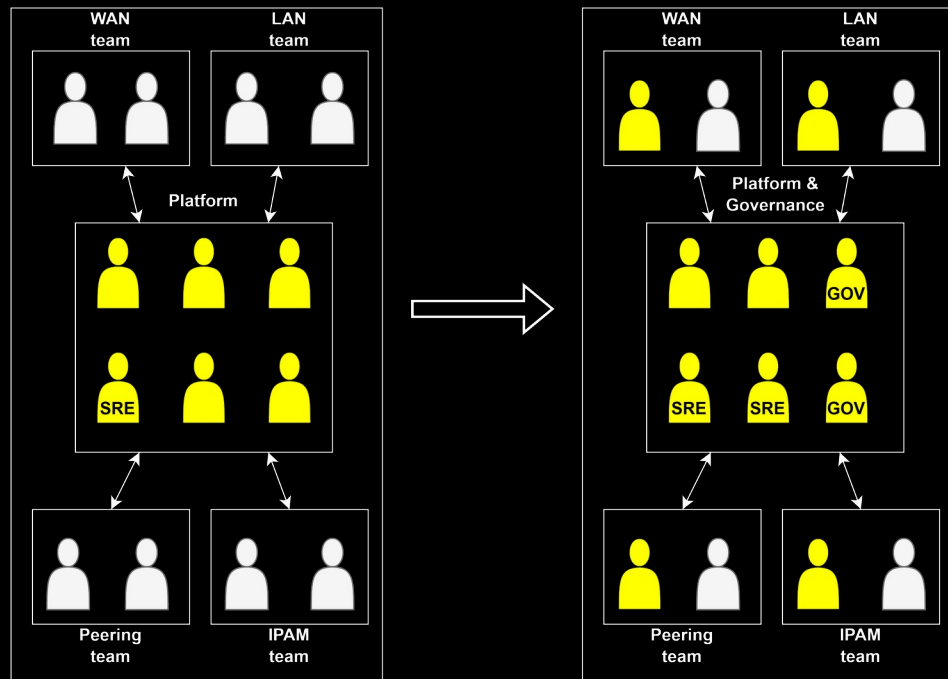
Staying on track - platform

Scaling out

2022

2024

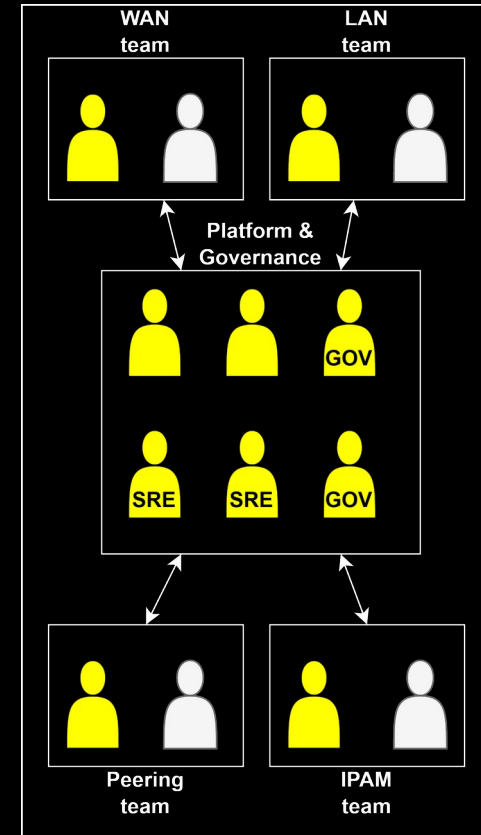
now



Fresh Look

"Organizations which design systems are constrained to produce designs which are copies of the communication structures of these organizations."

- Melvin E. Conway



Touch Points

- Weekly integration meetings
- Shared responsibility
 - Automation: platform, data syncs, etc.
 - Teams: their respective workflows
 - PRs: reviewed from both sides

Keep Customers Happy

Platform

- HTTP status codes
- Latency - % of requests > 3sec

Continuous pursuit of performance

Without Sacrificing Boundaries

Developer Experience

- Time to deploy
- Code test coverage
- CI/CD pipeline success rate

Continuous pursuit of **quality**

Refueling - conclusion

Lessons Learned

- Management buy-in
- Blameless culture
- Shared responsibility
- Focus on MVP
- Automation-friendly architecture

Next Steps

- We are nowhere near done
- Onboard more teams and workflows
- Working through performance

Conclusion

Why haven't we seen full adoption of network automation [in our organization], yet?

Conclusion

Why haven't we seen full adoption of network automation [in our organization], yet?

Too many cool things to do!



Questions?